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Subject: Cyber Range

Activity: Activity 02

GUI-Based Password Spraying Attack and Smart Account Lockout Defense Simulation

1. Introduction

Cybersecurity systems today face many authentication-based attacks. One of the most dangerous among them is the **Password Spraying Attack**, where an attacker tries a single common password against multiple user accounts to avoid detection. This project presents a **GUI-based Cyber Range simulation** that demonstrates how password spraying attacks occur and how a **Smart Account Lockout Defense** can detect and stop such attacks. The system is implemented in Python and provides a visual environment to understand both offensive and defensive cybersecurity operations. The project follows the concept of **Cyber Range as a Service (CRaaS)** by providing a controlled virtual environment where users can simulate attacks safely without affecting real systems.

2. Objective

The main objectives of this project are:

- To develop a **GUI-based cyber attack simulation tool**
- To simulate a **Password Spraying Attack**
- To implement a **Smart Lockout Defense**
- To calculate a **risk score**
- To display **real-time logs**
- To improve understanding of authentication security threats

3. Tools and Technologies Used

The following tools were used:

- **Python** – Core programming language
- **Tkinter** – GUI development
- **Datetime module** – Event timestamps
- **Random module** – Attack simulation
- **VS Code / IDLE** – Development environment

4. System Architecture

The system follows the architecture below:

User → GUI Interface → Attack Engine → Detection Module → Defense Engine → Result Display

Explanation:

- User enters the attack password
- Attack module simulates login attempts
- Detection module analyzes suspicious activity
- Defense system triggers account lockout
- Results are shown in the GUI window

5. Implementation

5.1 User Interface

The GUI includes:

- Password input field
- Start simulation button
- Risk score display
- Attack log window
- System status display
- Reset button

The interface is designed to be simple and user friendly.

5.2 Attack Module

The attack module performs:

- One common password
- Multiple user accounts
- Sequential login attempts
- Logging of failures and successes

Example:
Attacker tries:

Welcome123

Against users:

- alice
- bob
- admin
- john

This simulates a real password spraying scenario.

5.3 Detection Module

The detection engine checks:

- Same password used repeatedly
- Multiple accounts targeted
- Failure frequency
- Attack pattern

If suspicious behavior is found:

- The attack is classified as malicious
- Risk score increases
- Defense is activated

6. Risk Analysis

The system calculates a **Risk Score**.

Formula:

Risk score depends on:

- Failed attempts
- Number of users targeted
- Repeated password use

Risk Levels

Risk Score	Level
0–20	Low
21–40	Medium
41–60	High
61+	Critical

This helps identify the severity of the attack.

7. Defense Mechanism

The defense system uses:

Smart Account Lockout

When an attack is detected:

- Source is blocked temporarily
- Further attempts are denied
- Alert is generated
- System status changes

Defensive actions

- Login blocked

- Event logged
- User warned
- System secured

8. Working Process

The system works in the following steps:

1. User launches the GUI
2. Enter attack password
3. Click **Start Simulation**
4. Attack begins
5. Detection engine monitors behavior
6. Risk score is calculated
7. Defense activates
8. Output is displayed

9. Results

The project successfully demonstrates:

- Password spraying attack behavior
- Suspicious login detection
- Automatic account lockout
- Real-time monitoring
- Risk scoring

Observations:

- Repeated passwords increase risk
- Multiple account targeting triggers alert
- Smart defense blocks attack automatically

10. Advantages

Advantages of the system:

- Easy to understand
- Interactive GUI
- Safe simulation
- Useful for learning
- Real-time output
- Strong cyber range relevance

11. Limitations

Current limitations:

- Simulated environment only
- No real network connection
- Small user database
- Temporary lockout only
- No database integration

12. Conclusion

This project successfully demonstrates a **GUI-Based Password Spraying Attack and Smart Account Lockout Defense Simulation** in a Cyber Range environment.

It helps users understand:

- How password spraying works
- How attacks can be detected
- How smart defenses respond automatically

The project is useful as an educational cybersecurity tool for students and training environments.

13. Future Enhancements

Future improvements can include:

- Database integration
- User authentication
- Graphical analytics
- Real-time network monitoring
- Multi-user support
- Cloud-based deployment

14. References

- Tkinter Documentation- https://docs.python.org/3.10/library/tkinter.html?
- Cybersecurity Authentication Research Papers- https://owasp.org/www-community/attacks/Password_Spraying_Attack?
- Cyber Range Security Training Resources- https://cheatsheetseries.owasp.org/cheatsheets/Credential_Stuffing_Prevention_Cheat_Sheet.html?